README — Revision_Deflation_CJT_and_Kohn-Sham

PROJECT spacetimes: the 16-component Dirac field with mass term and self-interaction on the 4+4 and 7+1 spacetimes whose coordinates x5, x6, x7 deflate exponentially; Kohn–Sham and CJT analyses

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1. What this folder is

The folder holds one sub-project of PROJECT spacetimes. Its hypotheses are the author's:

- The three extra coordinates $x5$, $x6$, $x7$ deflate exponentially in the ordinary time $x4$. The metric is diagonal, and its components along them are $g_{tt}^{(4+4)} = -\text{const}_t^2\, e^{-2\,\text{rate}_t\, x4}$ on the 4+4 spacetime (where they are time-like) and $g_{tt}^{(7+1)} = +\text{const}_t^2\, e^{-2\,\text{rate}_t\, x4}$ on the 7+1 spacetime (where they are space-like), for $t = 5, 6, 7$, with the positive rates $\text{rate}_5 = \text{g55}$, $\text{rate}_6 = \text{g66}$, $\text{rate}_7 = \text{g77}$ and positive constants const_t.
- The Lagrangian of the sixteen-component Dirac field contains a kinetic term, a **mass term** and a **self-interaction** that is quartic in the field, with nine couplings $g_0, \ldots, g_8$.
- The consequences are worked out with two methods, the Kohn–Sham ansatz and the effective potential for composite operators of Cornwall, Jackiw and Tomboulis (CJT), each as a check of the other and of the earlier results.

Everything in the folder was computed for it: every number in a document comes from a data file that a command of the program wrote, and every table is inserted by a script.

2. Where to start reading

if you want	read
the physics of one spacetime, complete	the manuscript for that spacetime, in `manuscript/`; it defines every symbol it uses and contains its own appendices
the definitions: metric, Dirac matrices printed in full, Lagrangians, field equations, derived step by step for both spacetimes side by side	`dirac matrices deflating`
the meaning of a symbol	`provenance/P1_master_symbols_spacetimes` (every symbol of the project, with one meaning)
how a number was computed, or how to compute it again	`provenance/P2_calculations_code_and_checks`
what was asked, what was done, what was not, and which decisions are open	`provenance/P0_task_rules_deviations_decisions`

Each document exists as `.md`, `.tex` and `.pdf`, and each is complete in itself.

3. Contents

path	content
`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime`	manuscript for the 4+4 spacetime (`.tex`, `.pdf`, `.md`)
`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 7+1 Deflating Spacetime`	manuscript for the 7+1 spacetime
`manuscript/deflation_refs.bib`	the references of the manuscripts
`dirac matrices deflating`	geometry, Dirac matrices, Lagrangians, field equations, solutions; reports of the checks
`provenance/P0_task_rules_deviations_decisions`	the task, the rules, deviations, decisions, error log
`provenance/P1_master_symbols_spacetimes`	the MASTER registry of symbols and the proof that no symbol has two meanings
`provenance/P2_calculations_code_and_checks`	commands, instructions, data files, figures, complete output of all checks, source files
`provenance/symbols_master.csv`	the registry as a table (229 symbols)
`provenance/generated/`	tables and numbers written by the scripts from the data files
`code/deflation-cjt-ks-rs/`	the Rust program `deflation_cjt_ks` (33 source files, 15007 lines, 68 unit tests)
`code/scripts/`	independent verification, figures, and the scripts that build and check the documents
`data/`	111 data files and the reports of the checks
`figures/`	30 figures (`.pdf` and `.png`)
`docs_src/`	the sources of all documents

4. How to run everything

What is needed. A Rust toolchain (`cargo`); Python 3 with numpy, scipy, sympy, mpmath and matplotlib; for the documents, pandoc, XeLaTeX with latexmk, and pdflatex with REVTeX 4.2 and BibTeX. The program uses the Rust crate `code/spacetimes-rs` at the top of the repository as a library; nothing is downloaded.

Where. All commands are run from this folder. On the author's computer:

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation_CJT_and_Kohn-Sham"
```

Step 1. Compile the program and run its unit tests.

```
cargo test  --release --manifest-path code/deflation-cjt-ks-rs/Cargo.toml
cargo build --release --manifest-path code/deflation-cjt-ks-rs/Cargo.toml
```

The first command must end with `test result: ok`. The second builds the program `code/deflation-cjt-ks-rs/target/release/deflation_cjt_ks`.

Step 2. Run the commands of the program. Each writes its data files and its report of checks into `data` and ends with a line such as `kohnsham: 13 of 13 checks passed`.

```
B=code/deflation-cjt-ks-rs/target/release/deflation_cjt_ks
$B help
for c in selfcoupling kohnsham pairing gravity matrices modes symmetry channels \
         history quench paired cosmology matter background spectrum; do $B $c; done
$B forecast
$B superfluid
$B families
```

The first line abbreviates the path of the program, and `help` lists the commands with a description of each. The loop runs fifteen commands, each of which takes from a few seconds to about two minutes. The last three take longer (about a quarter of an hour, three quarters of an hour, and an hour and a quarter on a machine with two processors).

Step 3. The independent verifications.

```
python3 code/scripts/verify_deflating_independent.py
python3 code/scripts/verify_cjt_ks_independent.py
python3 code/scripts/mutation_test_independent.py
python3 code/scripts/vendor_eb.py --check
```

The first two repeat the central results with computer algebra and general-purpose numerical tools (43 of 43 and 37 of 37 checks pass). The third introduces deliberate errors into copies of the two scripts, one at a time, and records whether the checks notice (36 of 36 detected; about six minutes). The fourth verifies that four modules of the program are exact copies of modules of the library crate with a listed set of changes.

Step 4. Figures, tables, documents, and the checks of the documents.

```
python3 code/scripts/make_figures.py
python3 code/scripts/write_symbol_registry.py
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py
python3 code/scripts/build_manuscripts.py
bash    code/scripts/build_docs.sh
bash    code/scripts/check_all.sh
```

In this order: draw every figure from the data files; write the registry of symbols; write every table and every computed number of the documents from the data files; assemble the Markdown documents; assemble and typeset the two manuscripts; typeset the other documents; run the checks of the documents and read the verdict of every report. The last command prints one line for each check and must end with `ALL PASS`.

5. The checks, in numbers

what	result
checks of the program (exact where possible; reports `data/*_checks.txt`)	295 of 295 pass
unit tests of the program	68 pass
independent verification, geometry, Dirac field, modes, cosmology	43 of 43 pass
independent verification, self-interaction, Kohn–Sham, CJT, pairing, gravity	37 of 37 pass
deliberate errors introduced into the independent scripts	36 of 36 detected
checks of the registry of symbols, of the documents and of the data files against it	7 of 7 pass
numbers of the running text traced to a statement of a program	500 numbers in 26 documents, 0 not traced

6. Conventions in one place

- Coordinates $x0, \ldots, x7$: $x1, x2, x3$ ordinary space, $x4$ ordinary time, $x0$ a hidden space-like coordinate, $x5, x6, x7$ the deflating coordinates.
- Sign of the metric: space-like components positive, time-like components negative.
- The rates g55, g66, g77 are written upright and never with a subscript; a component of the metric carries the superscript $(4+4)$ or $(7+1)$; a coupling of the self-interaction, $g_0, \ldots, g_8$, carries one digit as its subscript.
- Units: $\hbar = c = 1$.
- In the output of the programs, “the main project” means the part of PROJECT spacetimes that treats the flat 4+4 and 7+1 spacetimes; its program is used as a library, and several checks compare with its data files.

7. What is open

Three matters need the author’s decision: the form of the self-interaction on the 4+4 spacetime (two forms are computed; one is adopted), the role of a bare mass beside the Vorton-Scalar, and the reading of the couplings of the reference example. They are set out, with the deviations from the instructions and the log of errors, in the document P0.